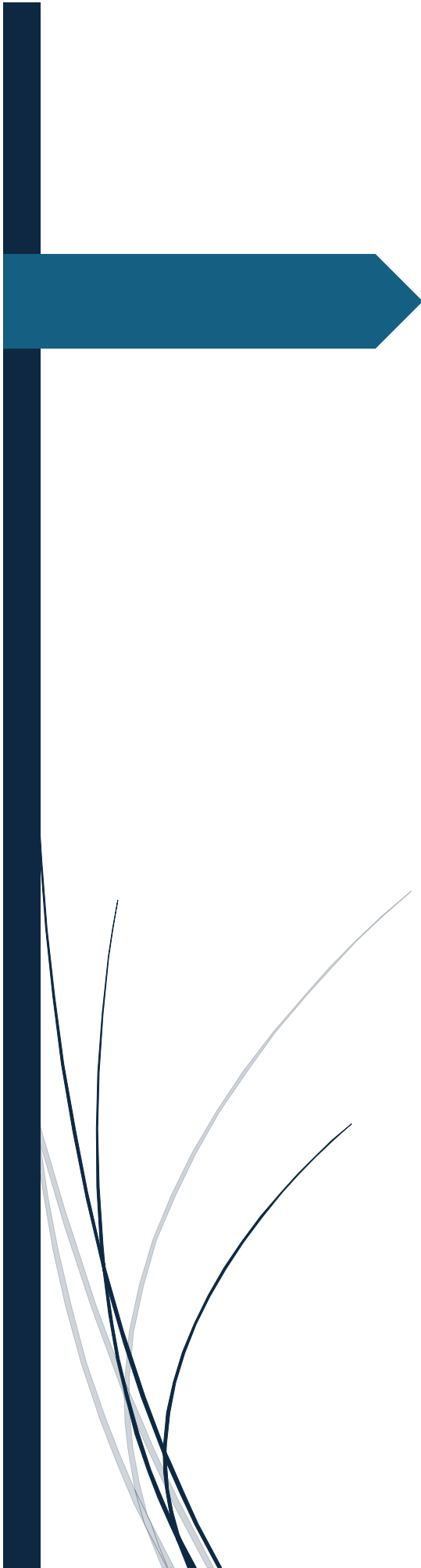




Assignment 3

BIOS & PORT Troubleshooting
Theory

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

Error message	Usually means
1. No Boot Device Found	The HDD/SSD is not detected or has a boot problem.
2. CMOS Checksum Error	BIOS settings may be damaged, or the CMOS battery may be weak.
3. Keyboard Error	The keyboard is not connected or is not working.
4. Memory Error / RAM Error	The RAM may be loose, damaged, or not detected.
5. CPU Fan Error	The CPU fan may not be connected or may not be working properly.

2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

❖ Problem:

When the PC starts:



- It gives **3 beeps repeatedly**
- Nothing appears on the screen

➤ Meaning in AMI BIOS

3 beeps usually mean a RAM (memory) problem.

The BIOS cannot properly detect or access the RAM.

- **Troubleshooting Steps:**

Step 1: Check the RAM

- Turn OFF the PC.
- Remove the RAM stick.
- Clean the RAM contacts gently.
- Put the RAM back properly.
- Turn ON the PC and check.

Step 2: Try another RAM slot

- Turn OFF the PC.
- Move the RAM to another RAM slot.
- Turn ON the PC again.
- If possible, test with another working RAM stick.

3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.



- **Motherboard Manufacturer: ASUS**
- **BIOS Version: 316**

4. You are given a screenshot of a POST error message that says '**CMOS checksum error – Defaults loaded.**' Explain what this means and describe how you would resolve this issue in a real system. Hint: Think about the **CMOS battery** and **BIOS settings reset**.
gimme step by step

Why it happens:

- The **CMOS battery** is weak/dead.
- BIOS settings are **lost or corrupted**.
- The computer detects that the stored BIOS settings are incorrect, so it **loads default settings**.

How to fix it:

1. **Replace the CMOS battery** with a new one.
2. Start the computer and enter **BIOS/UEFI**.
3. Set the **correct date and time**.
4. Check important BIOS settings, such as **boot order**.
5. **Save and Exit BIOS**.
6. Restart the computer and check if the error is gone.